WORKED SOLUTION

STEP I 1996, Question 14

Probability and Statistics • Runs and Conditional Probability

> **STEP I 1996, Question 14**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/96-s1-q14

The key idea

Condition on how the current run ends. Once the coin changes face, everything that happened before is irrelevant: the old run is broken, and the new run has length one. So after a head the game "restarts" the moment a tail appears, in exactly the state it would be in if the very first toss had been a tail, and vice versa. That gives two linear equations in two unknowns.

Solution

Write $x = \mathrm{P}(A \mid H)$ and $y = \mathrm{P}(A \mid T)$. Tosses are independent throughout. We take $r, s \geq 1$.

The equation for $\mathrm{P}(A \mid H)$

Suppose the first toss is a head. We now have a run of one head, and no run of tails has started. Look at the next $r-1$ tosses (tosses 2 to r).

- If all of them are heads, which has probability p^{r-1}, a run of r heads is completed at toss r. No tail has appeared at all, so certainly no run of s tails came first, and A occurs.
- Otherwise, which has probability $1 - p^{r-1}$, a tail appears among tosses $2, \ldots, r$ before the run of heads reaches length r. At the first such tail, the run of heads is broken without reaching r, and no earlier tail has appeared, so no run of s tails has happened yet either. We are now holding a run of exactly one tail, and nothing earlier can contribute to any future run.

> **Ask yourself** Why is the probability of A from that tail onwards exactly $\mathrm{P}(A \mid T)$?

After the first toss is a tail, the situation is: a run of one tail, no run of heads in progress,

and the future tosses independent of the past. That is precisely the situation at the first tail in the second case above. Since later tosses are independent of earlier ones, and which of the two runs comes first depends only on the current run and the future tosses, the conditional probability of A from that point is y. Hence, by the law of total probability,

$$\mathrm{P}(A \mid H) = p^{r-1} + \left(1 - p^{r-1}\right)\mathrm{P}(A \mid T),$$

so $\alpha = r - 1$.

The equation for $\mathrm{P}(A \mid T)$

Suppose the first toss is a tail. Look at tosses 2 to s.

- If all are tails (probability q^{s-1}), a run of s tails is completed before any head appears, so A fails.
- Otherwise (probability $1 - q^{s-1}$), a head appears before the tails reach s in a row. At that head we have a run of one head and nothing in progress for tails, which is exactly the situation after a first-toss head. So from there A occurs with probability x.

Therefore

$$\mathrm{P}(A \mid T) = \left(1 - q^{s-1}\right)\mathrm{P}(A \mid H),$$

so $\beta = s - 1$.

(The edge cases fit too. If $r = 1$ the first equation says $x = 1$, which is right because a single head is already a run of length r. If $s = 1$ the second says $y = 0$.)

Solving

Substitute the second equation into the first:

$$x = p^{r-1} + \left(1 - p^{r-1}\right)\left(1 - q^{s-1}\right)x.$$

Now

$$1 - \left(1 - p^{r-1}\right)\left(1 - q^{s-1}\right) = p^{r-1} + q^{s-1} - p^{r-1}q^{s-1},$$

which is positive (it is at least $p^{r-1} > 0$ when $0 < p < 1$, since $q^{s-1}(1 - p^{r-1}) \geq 0$). So we may divide by it. Writing $D = p^{r-1} + q^{s-1} - p^{r-1}q^{s-1}$,

$$\mathrm{P}(A \mid H) = \frac{p^{r-1}}{D}, \qquad \mathrm{P}(A \mid T) = \frac{p^{r-1}\left(1 - q^{s-1}\right)}{D}.$$

Finally, by the law of total probability with $\mathrm{P}(H) = p$ and $\mathrm{P}(T) = q$:

$$\begin{aligned}\mathrm{P}(A) &= p\,\mathrm{P}(A \mid H) + q\,\mathrm{P}(A \mid T)\\ &= \frac{p^{r-1}\left(p + q - q^{s}\right)}{D}\\ &= \frac{p^{r-1}\left(1 - q^{s}\right)}{p^{r-1} + q^{s-1} - p^{r-1}q^{s-1}},\end{aligned}$$

using $p + q = 1$.

Checks

- Symmetry: for a fair coin with $r = s$, the answer should be $\frac{1}{2}$. With $p = q = \frac{1}{2}$ and $r = s = n$, the numerator is $2^{1-n}(1 - 2^{-n})$ and the denominator is $2^{2-n} - 2^{2-2n} = 2 \cdot 2^{1-n}(1 - 2^{-n})$, giving $\frac{1}{2}$.
- Small case: $r = 1$ gives $\mathrm{P}(A) = 1 - q^s$, the chance that the first s tosses are not all tails, which is correct.
- For $p = \frac{1}{2}$, $r = 3$, $s = 2$ the formulae give $\mathrm{P}(A \mid H) = \frac{2}{5}$, $\mathrm{P}(A \mid T) = \frac{1}{5}$ and $\mathrm{P}(A) = \frac{3}{10}$. We confirmed this by solving the full Markov chain exactly, and a simulation of 200 000 games gave 0.298.

Common slips

- Using p^r instead of p^{r-1}: the first head has already been given, so only $r - 1$ more are needed.
- Forgetting that $\mathrm{P}(A)$ needs the final weighting by p and q. The conditional probabilities alone are not the answer.

Beyond STEP

The argument is a *first-step analysis* of a Markov chain: the states are "current face and run length", and the restart property is the Markov property. The same method finds absorption probabilities and expected waiting times for any pattern of heads and tails, and leads on to Penney's game, where some patterns surprisingly beat others. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and first-step analysis at the right level.

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